Jia Li

Data Engineer · ETL / ELT & Data Pipelines · Data Science & Analytics Infrastructure

iammarine30@gmail.com | Beijing, China | Open to relocation (Japan / APAC) | English (Fluent)
Japanese (Elementary, studying) | Chinese (Native)



Summary

Data engineer with 10+ years building large-scale ETL/ELT pipelines, data platforms, and the analytics/ML infrastructure that sits downstream of them. At Alibaba, owns backend data pipeline infrastructure for an instant-commerce product processing **10M+ orders/day**, engineering high-throughput distributed ETL that parses and structures tens of billions of raw event records daily, and building automated data-quality/drift monitoring that other teams depend on operationally. Comfortable moving between core pipeline engineering and the analytics/data-science layer built on top of it — metric design, statistical anomaly detection, experimentation infrastructure, and production ML pipeline integration. Experienced mentoring engineers and leading teams of 5–17.

Core Competencies

ETL / ELT & Pipeline Engineering	High-throughput ingest pipelines parsing tens of billions of raw log/event records/day into structured, analytics-ready tables · multi-source reconciliation across independent upstream systems · ETL/ELT design at 10M+ daily-event scale · Regex/multi-condition parsing, cycle-removal path reconstruction
Data Platform & Warehouse	Distributed pipeline architecture on MaxCompute, Alibaba's large-scale distributed data warehouse · custom stateful UDAFs executing in cluster memory to eliminate GROUP BY shuffle overhead on high-volume aggregations · warehouse table design and query optimization
Data Science & Analytics	Metric & KPI framework design (60+ indicator reporting platform) · statistical anomaly and rolling-window drift detection · root-cause analysis pipelines · production ML pipeline integration (PyTorch embedding models in recommendation infrastructure) · experimentation-adjacent reporting infrastructure
Reliability & Observability	Automated data-quality and drift monitoring: SLA freshness, completeness, cross-source reconciliation · production diagnostic dashboards · on-call incident diagnosis and root-cause routing via automated anomaly-detection pipelines
Production Ownership	10+ years owning production data infrastructure at hyperscale on Alibaba and JD.com internal platforms · end-to-end responsibility for pipeline design, monitoring, on-call, and incident response
Scale Handled	10M+ daily orders (Taobao Flash Sale) · 502,000 orders/24h zero-downtime record (JD.com) · tens of billions of log/event records processed into structured pipelines daily
Languages & Tools	SQL (expert) · Java · Python · NumPy / Pandas · MaxCompute · PyTorch (production pipeline integration) · Git
Mentorship & Cross-Team Collaboration	Led and mentored engineering teams of 5–17 · built data contracts and platform services consumed directly by data engineers, BI, and algorithm teams across independent business units · on-site technical command during peak-load operational events

Experience

Alibaba Group — 2018 – Present

Senior Data R&D Expert (P8) — Taobao Flash Sale — 2021 – Present

Own backend ETL/ELT and data platform infrastructure for an instant-commerce product processing 10M+ orders/day, providing pipelines, APIs, and shared data assets consumed by product, BI, algorithm, and data science teams.

- **High-throughput ETL.** Engineered MaxCompute pipelines with custom stateful UDAFs executing in cluster memory, eliminating SQL GROUP BY shuffle overhead on high-volume aggregations — cutting both runtime and compute cost at hyperscale.
- **Data reliability & observability.** Built automated data-quality and drift monitoring covering SLA freshness, completeness, cross-source reconciliation, and rolling-window statistical drift detection, feeding production diagnostic dashboards used by on-call engineers.
- **Automated anomaly diagnosis (data science).** Built an autonomous pipeline (anomaly detection → dynamic multi-dimensional query assembly → statistical analysis → owner routing) that diagnoses business-metric anomalies end-to-end and routes to the responsible team with evidence attached.
- **Platform services for downstream teams.** Designed data contracts and APIs consumed by product, BI, and algorithm teams — the platform layer other engineering teams build on top of.
- **ML infrastructure integration.** Deployed PyTorch multi-modal embedding pipelines into production recommendation infrastructure, handling batch inference at scale.

Data R&D Expert (P7) — Taobao & Tmall Data Mid-Platform — 2020 – 2021

- **Multi-tenant ETL platform.** Built shared data-ingest infrastructure spanning four independent business units with different source systems — conformed-dimension modeling instead of a forked pipeline per tenant, serving a 60+ indicator reporting framework through common infrastructure.
- **Log-to-table pipeline at scale.** Processed tens of billions of raw behavior-log records into structured, analytics-ready tables via Regex/multi-condition parsing and cycle-removal path reconstruction. Became shared infrastructure adopted by downstream analytics teams.

Data Product Expert (P7) — Freshippo (Alibaba O2O Retail) 2018 – 2020

- Formed and led a 5-engineer team building real-time ETL/data-ingest infrastructure and internal APIs supporting network growth from 90 to 300 stores.
- Designed a predictive inventory-forecasting pipeline operated by frontline logistics teams, supporting a single campaign that cleared 100M+ units without infrastructure incident.

JD.com Group 2012 – 2018

Product R&D Manager · WMS R&D Manager — Logistics & Retail Technology

- Led and mentored a 17-engineer team building the backend infrastructure and warehouse-management platform for **Asia No.1**, then Asia's largest automated fulfillment facility. On-site infrastructure command lead during Double 11 2015: **502,000 outbound orders in 24 hours with zero downtime** — a single-warehouse world record at the time.
- Led and mentored a 6-engineer team building RFID-based automated identification pipelines and dynamic replenishment infrastructure for unmanned-retail smart logistics.
- Built Regex-based structured log-parsing ETL infrastructure converting raw operational log streams into analytics-ready tables — adopted as shared infrastructure by BI and analytics teams.

Earlier — Telecom Software Engineering (NSN, ZTE Soft) 2006 – 2012

- Built enterprise OSS/BSS backend infrastructure in Java for China Mobile (Beijing) and PCCW (Hong Kong). Five years on-site in Hong Kong in a fully English-speaking professional environment.

Education & Additional Information

Henan University Bachelor of Economics

Languages: Chinese (native) · English (fluent — five years working on-site in Hong Kong) · Japanese (elementary, currently studying)

Work authorization: Eligible for the Japan Engineer / Specialist in Humanities / International Services visa under the 10-year practical experience provision.